

Diurnal and seasonal trends and associated determinants of surface urban heat islands in large Bangladesh cities

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ABSTRACT

Many surface urban heat island (SUHI) studies have been conducted around the globe, however there is still a lack of information available regarding the intensity of SUHI (SUHII) in Bangladesh cities. This study focused on diurnal and seasonal SUHI variability, temporal trends and possible drivers in five major cities. Mean annual daytime SUHII ranged from 2.88 °C for Dhaka to 0.84 °C for Rajshahi, while nighttime intensity varied from 1.91 °C (Chittagong) to 0.30 °C (Sylhet). The pre-monsoon period exhibited the greatest magnitude and the seasonal amplitude during the winter season was positive for Dhaka and Khulna but negative for the other cities. Correlation analysis indicated that a dense city population, a high degree of imperviousness and the absence of greenery were likely to act singly, or in combination, to increase urban warming within these cities. An increasing warming trend during daytime was observed. The urban population of Bangladesh is projected to increase substantially in future (i.e., to 81.4 million by 2029), so the findings of this study provide valuable insights into this warming issue and will assist in the development of effective local-scale climate change adaptation plans.

1. Introduction

Globally, the amount of land used for urban expansion is increasing at a greater rate than the urban population (Huang et al., 2021), resulting in vast areas of natural land cover being removed and replaced with impervious surfaces (Gong et al., 2020; Wentz et al., 2008). This permanent change in the character of the land surface has resulted in a number of environmental impacts, the most notable of which is the increased warming of urban areas (Wang et al., 2020; Voogt & Oke, 2003). Many cities now experience elevated temperatures relative to their surroundings – a phenomenon known as the urban heat island (UHI) (Roth, 2007; Arnfield, 2003; Oke, 1973). Three types of UHI are currently recognised; i) the surface urban heat island (SUHI); ii) the canopy layer heat island (CUHI); and iii) the boundary layer heat island (BUHI) (Voogt & Oke, 2003). This study focuses on the SUHI.

Thermal infrared data obtained from earth-observing satellites is commonly used to examine the spatial patterns, temporal dynamics and causal factors of SUHI globally (Li et al., 2020; Manoli et al., 2019; Chakraborty & Lee, 2019; Clinton & Gong, 2013; Peng et al., 2012), regionally (Raj et al., 2020; Shastri et al., 2017; Zhou et al., 2017; Yao et al., 2017; Imhoff et al., 2010) and locally (Hu et al., 2019; Chun &

Guldmann, 2018; Huang et al., 2016; Roy & Singh, 2015; Quan et al., 2014; Tomlinson et al., 2012; Yuan & Bauer, 2007). Although these studies have increased our knowledge regarding spatiotemporal dynamics, the use of diverse data sources, quantification methods and differing urban boundary definitions can introduce uncertainty (Manoli et al., 2019), particularly when studying dense and rapidly growing cities. Existing studies on Bangladesh cities are based on limited time-frame data (only daytime using selected years), scope (single city or a few cities) or have examined only the impact of land use/cover change on urban land surface temperature (LST) (Dewan & Corner, 2012; Kant et al., 2018; Roy et al., 2020; Dewan & Corner, 2014). Three studies have investigated SUHI only for a single season, either for winter or summer/monsoon (Itzhak-Ben-Shalom et al., 2017; Kant et al., 2018; Roy et al., 2020). These studies however have failed to provide detailed diurnal and seasonal SUHI patterns, and determinants, for Bangladesh cities.

Climatic zones, and the individual features of the cities found within those zones, can greatly influence SUHI intensity (Zhou et al., 2017; Mirzaei & Haghighat, 2010). Population, for example, showed a negligible relationship with SUHI at the global scale (Clinton & Gong, 2013; Peng et al., 2012). Regional and local studies have, however, suggested

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some correlation (Raj et al., 2020; Zhou et al., 2018; Hung et al., 2006) owing to the fact that the population of an area both directly, as well as indirectly, influences urban climate (Rizwan et al., 2008). Similarly, the effect of waterbodies was shown to vary according to the makeup of the cities examined (Song & Wu, 2016; Steeneveld et al., 2014). Other researchers have also noted the contribution of various factors to the variability of SUHI both at diurnal and seasonal scales (Sun et al., 2019; Chun & Guldmann, 2018; He, 2018; Chen et al., 2017). Since ecological settings, as well as the spatial layout of cities, vary considerably (Alexander, 2021; Zhang et al., 2010), it is difficult to determine factors controlling SUHI intensity when using global studies to examine local scale characteristics. Leaf presence/absence, and/or the types of vegetation or agricultural activities being conducted adjacent to a city, can also have a significant influence on urban warming (Manoli et al., 2019; Yao et al., 2018; Quan et al., 2016).

With an estimated population of 161 million (BBS, 2017), Bangladesh is currently experiencing rapid changes in the existing land use/cover with massive conversion of arable land and forest cover to urban surface (Xu et al., 2020). Since urban expansion is mostly unplanned in the country, many large cities are facing a myriad of environmental challenges, most notably the warming of urban areas. Studies utilising remote sensing imagery and downscaled climate models have indicated a rise in urban temperature over time in many cities (Chakraborty & Lee, 2019; Kant et al., 2018; Mourshed, 2011). Urban-area induced climatic modifications due to increasing human activities

have the potential to impact millions of urban inhabitants directly (e.g., mortality) or indirectly (e.g., loss of livelihoods), so a clear understanding of the underlying causes of local warming is required to assist in formulating location-specific adaptation and mitigation strategies.

The purposes of this study are to: (i) analyse diurnal and seasonal variations of SUHI; (ii) investigate the relationship between SUHI intensity with biophysical and socioeconomic descriptors; and (iii) examine temporal trends for the period 2003–2019.

2. Materials and methods

2.1. Study cities

Bangladesh (Fig. 1) is one of the most populous countries in the world with a density of 1,265 persons per km² (<https://worldpopulationreview.com/countries/bangladesh-population>). It is generally low-lying with most of the topographical surface seldom rising more than 10 m above mean sea level. The country has four distinct seasons: pre-monsoon (March–May), monsoon (June–September), post-monsoon (October–November) and winter (December–February). The climate is subtropical in nature and is characterised by cool and dry winters and hot humid summers. The rainy season exhibits marked seasonality in both rainfall and temperature.

The study focuses on five major cities - Dhaka, Chittagong, Khulna, Rajshahi and Sylhet, which have been selected based on population size

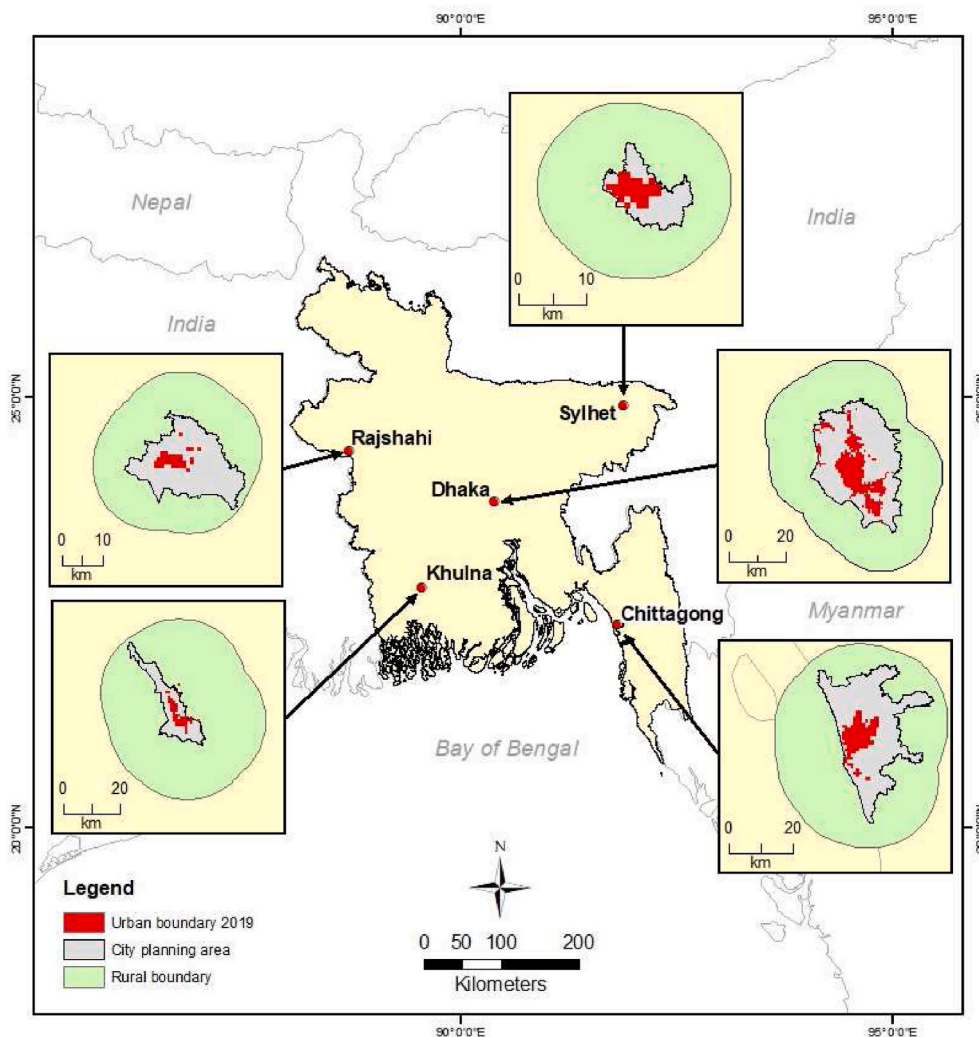


Fig. 1. Location of Bangladesh and five major cities.

(BBS, 2012), spatial attributes such as urban growth rate, ratio of built-up and non-built-up areas (Corner et al., 2014; Rahman et al., 2019) and the availability of ancillary data. Dhaka and Chittagong are the two largest urban agglomerations. The combined total population of these cities is 16.5 million, with Dhaka having the greatest contribution (10.4 m) (<https://worldpopulationreview.com/countries/cities/bangladesh>). Dhaka, Rajshahi and Sylhet are situated inland, while Khulna and Chittagong are located on the coast.

2.2. Data

This study uses version 6, Moderate Resolution Imaging Spectroradiometer (MODIS) products derived from the Aqua and Terra satellites, as well as a Landsat population product and Nighttime lights (NTL) data. Planning boundaries have been obtained from the development authorities of the respective cities. Data details are shown in Table 1.

The local overpass times of the MODIS satellites are nominally 0130 and 1030 local solar time (for Aqua) and 1330 and 2230 (for Terra). Calibration showed that MODIS land surface temperature product (LST) v6 is more accurate than previous versions with a bias of <0.5 K (Wan, 2014). Only clear sky pixels with a LST error of ≤ 2 K were considered. The 8-day product was aggregated to monthly to produce mean diurnal, annual and seasonal datasets for analysis.

The MOD09A1 product was used to compute various indices including the fraction of green vegetation (FVG) (Gutman & Ignatov, 1998), index-based built-up index (IBI) (Xu, 2008), and normalised difference water index (NDWI) (McFeeters, 1996). FVG was used instead of other popular vegetation indices (e.g., normalised difference vegetation index (NDVI)) as FVG is sensitive to the intensity of SUHI (Liu et al., 2019). Anthropogenic forcing, defined by annual nighttime lights (NTL), underwent geometric correction as well as inter- and cross-sensor calibration (see Dewan et al., 2021 for detail). A total of seven causative factors, namely aerosol optical depth (AOD), black sky albedo (BSA), FVG, IBI, population, NDWI and NTL, were generated. The grid size of the differing datasets varied, so a nominal 1 km spatial resolution was used.

2.3. Analytical techniques

The literature indicates that a number of methods can be used to define the SUHI of a city. A large part of the methodological differences relates to the manner in which the distance from the urban core (used for calculating intensity) is defined (Clinton & Gong, 2013). A comparison of five well-known methods indicated that using adjacent suburban areas as a reference tended to underestimate SUHI, while ignoring the influence of waterbodies tended to overestimate SUHI (Yao et al., 2018). In this study, annual MODIS land use and land cover (LULC) data from 2003 to 2019 was used to delineate the urban and rural boundaries; this uses the International Geosphere-Biosphere Programme (IGBP) classification scheme (Sulla-Menashe & Friedl, 2018). Built-up area intensity was extracted from timeseries LULC data and only those with high intensity built-up pixels were considered as urban boundary (Zhou et al.,

2014). The rural boundary delineation for each year, and for each city, was an iterative process. Rural sites were defined at a substantial distance from these identified urban areas. It is important to note that the buffer distance was created dynamically for every year, utilising annual LULC data. The buffer distance ranged between 15 and 20 km, based on the LULC-defined urban boundaries of the cities (Fig. 1). Water pixels were removed from urban areas, and built-up pixels (if present) were also removed from the rural locations. The final urban and rural site areas were comparable in size. As noted earlier, diurnally and seasonally aggregated LST data, averaged over the study period (2003–2019), were used to compute SUHI intensity (SUHI). Temporal SUHI refers to the difference in mean LST values while the spatial pattern is defined as the pixel-wise difference in mean LST between urban and rural areas for each city.

Seven different variables were extracted from the data, each representing a specific factor of interest showing the difference between the urban and rural areas (referred to as delta ' Δ '). These variables were greenness (Δ FVG), black sky albedo (Δ BSA), aerosol (Δ AOD), impervious surface (Δ IBI), nighttime lights (Δ NTL), population (Δ POP) and waterbodies (Δ WI). While five of the variables had available seasonal information, both Δ NTL and Δ POP only had annual data. The Pearson's correlation coefficients, between the diurnal and seasonal SUHI and the seven variables, were calculated to allow an assessment of factors with the potential to impact urban temperature. Tests for significance at the 95% and 99% levels were considered.

A non-parametric Mann-Kendall (MK) test (Kendall, 1948; Mann, 1945) was used to determine whether the SUHI in the cities was increasing or decreasing over time. The slope was estimated using the Theil-Sen slope (Sen, 1968) with a positive slope representing an increasing trend and a negative slope denoting a decreasing trend. The trend was tested at both diurnal and seasonal scales. Fig. 2 provides a methodological flow of the work undertaken.

3. Results

3.1. Diurnal variability

The spatial distribution of SUHI for the satellite overpass times, averaged over the period 2003–2019, is shown in Fig. 3a–d. This indicates that mean SUHI is positive for the five cities during all four-satellite data-capturing time periods, particularly morning hour SUHI (1030, Fig. 3a) varies from 2.84 °C (Dhaka) to 1.97 °C (Chittagong), 1.30 °C (Khulna), 0.78 °C (Rajshahi) and 1.10 °C (Sylhet). A slight increase in SUHI during the afternoon hours (1330, Fig. 3b) can be seen in four cities (2.93 °C for Dhaka, 2.12 °C for Chittagong, 0.85 °C for Khulna and 0.90 °C for Rajshahi), while in Sylhet it decreases to 0.65 °C. Dhaka exhibits 1.59 °C (2230) and 1.52 °C (0130) SUHI while Chittagong shows 1.94 °C (2230) and 1.88 °C (0130) (Fig. 3c–d).

An analysis of the Terra and Aqua data indicates that the day and nighttime SUHI from the Aqua satellite is greater for three cities, including for Dhaka and Chittagong. The values for Sylhet and Khulna, however, are reversed (Supplementary Fig. S1). The spatial pattern of

Table 1
Summary of data used in this work.

Data/Product	Type	Spatial Resolution	From-to	Temporal resolution	Reference
MYD11A2	LST: V6, 8-day, Level 3	1 km	2003–2019	8-day	https://doi.org/10.5067/MODIS/MYD11A2.006
MOD11A2	LST: V6, 8-day, Level 3	1 km	2003–2019	8-day	https://doi.org/10.5067/MODIS/MOD11A2.006
MCD12Q1 (LULC)	LULC	1 km	2003–2019	yearly	https://doi.org/10.5067/MODIS/MCD12Q1.006
MCD19A1	Aerosol	1 km	2003–2019	daily	https://doi.org/10.5067/MODIS/MCD19A1.006
MCD43A3	Short wave, black sky albedo	500 m	2003–2019	daily	https://doi.org/10.5067/MODIS/MCD43A3.006
MOD09A1	Surface reflectance	500 m	2003–2019	8-day	https://doi.org/10.5067/MODIS/MOD09A1.006
Landsat	24-hr mobile population	1 km	2003–2019	yearly	https://landsat.ornl.gov/
DMSP-OLS	Nighttime lights (NTL)	1 km	2003–2013	yearly	https://eogdata.mines.edu/download_dnb_composites.html
VIIRS	Nighttime lights (NTL)	1 km	2013–2019	Monthly	https://ngdc.noaa.gov/eog/download.html
Planning boundaries	Shape file	–	2019	–	City planning authority

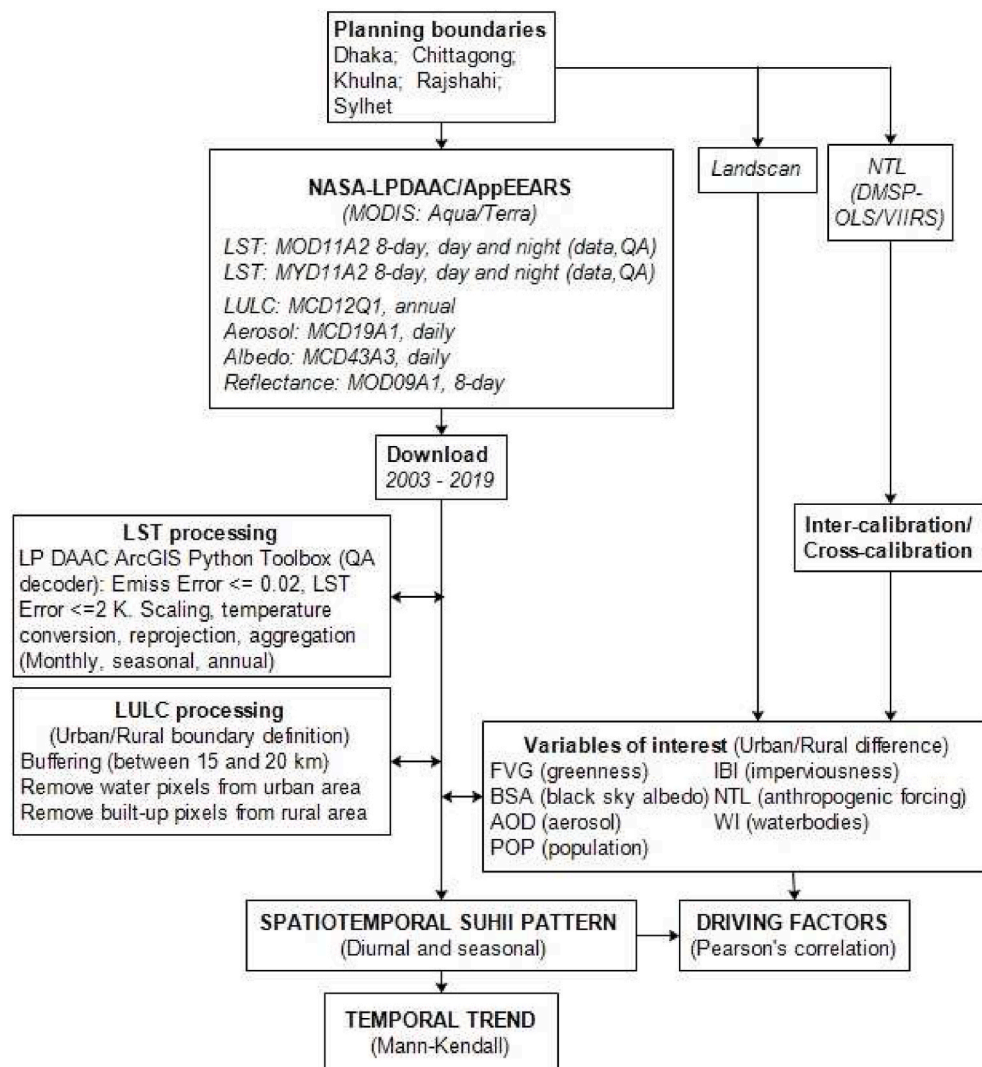


Fig. 2. Flowchart, showing the sequence of steps used to estimate surface urban heat island intensity (SUHII), determinants and trends for each city.

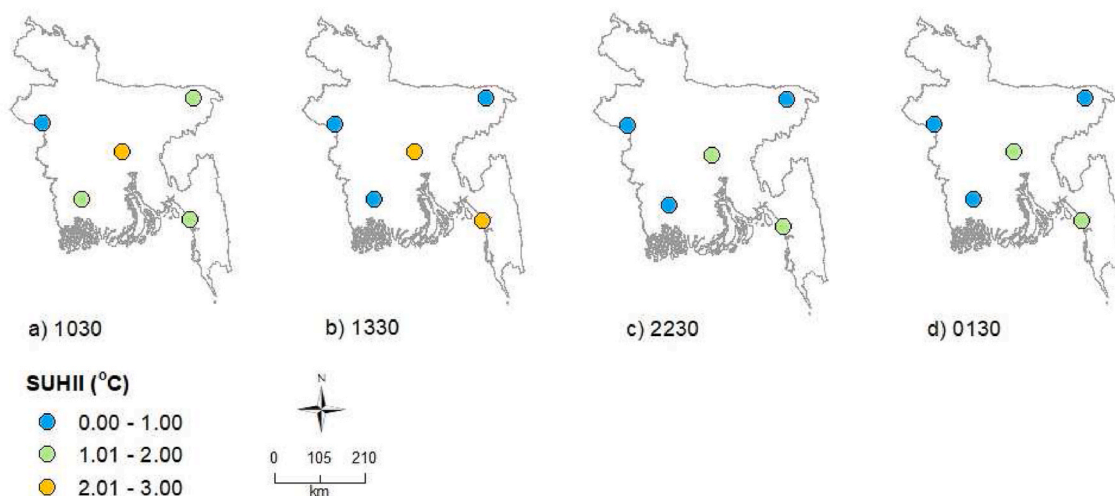


Fig. 3. Spatial distribution of SUHII during Terra and Aqua overpass times. The intensity is averaged over the period 2003–2019 for each city.

SUHII in individual cities is shown in [Supplementary Fig. S2](#). Day and night variability, obtained from the combined data from both sensors is shown in [Supplementary Fig. S3](#), which shows annual daytime SUHII is

greatest for Dhaka and nighttime for Chittagong. The mean SUHII amplitude (the difference between day and nighttime SUHII) is positive for all cities but the magnitude varies.

3.2. Seasonal variations

Daytime intensity in the pre-monsoon period is greater than during the night across all seasons (Table 2). For example, among four overpass times, the SUHII is highest at 1330 for Dhaka (3.74 °C), followed by Chittagong (2.14 °C). Nighttime SUHI intensity recorded by the Terra satellite is also greater for Dhaka (1.63 °C) and Chittagong (1.58 °C) at 2230. During winter, however, Dhaka and Chittagong exhibit the greatest intensity both during the day and at night (Table 2).

The seasonal day and nighttime SUHII varies greatly as shown in Fig. 4(a–h). During the pre-monsoon season, mean daytime SUHII ranges from 1.01 °C (Khulna) to 3.44 °C (Dhaka) while at nighttime it varies from 0.49 °C (Sylhet) to 1.51 °C (Chittagong) (Fig. 4a–b). Although the magnitude of SUHII tends to reduce significantly during the monsoon night, daytime SUHII ranges from 1.03 °C (Khulna) to 3.03 °C (Dhaka) (Fig. 4c). Conversely, a negative SUHII can be observed in Khulna (−0.23 °C) and Sylhet (−0.84 °C) during the monsoon night (Fig. 4d). A similar pattern is seen during the post-monsoon season where all cities record positive daytime SUHII (Fig. 4e), while at night Khulna and Sylhet reveal negative intensity (−0.07 °C and −0.02 °C, respectively) (Fig. 4f). Positive SUHII can be observed in both day and nighttime across all cities during the winter season (Fig. 4g–h).

The spatial distribution of day and night SUHII in different seasons is presented in Supplementary Fig. S4 and Fig. S5, and the average day, night and day-night variability of SUHII is shown in Supplementary Fig. S6.

3.3. Relationship between SUHII and causal factors

Correlations between day and nighttime SUHII and any potential drivers have been calculated at both diurnal and seasonal scales. Daytime SUHII is significantly and positively correlated with imperviousness (defined by the Δ IBI), at the diurnal scale (Table 3). Similarly, Δ POP plays a significant role in affecting the daytime intensity of SUHII in Dhaka ($r = 0.935$, $p < 0.01$), Chittagong ($r = 0.784$, $p < 0.01$), Rajshahi ($r = 0.599$, $p < 0.01$) and Sylhet ($r = 0.486$, $p < 0.05$), however the relationship between these two variables for Khulna is weak. The Δ FVG displays a consistent, statistically negative correlation for daytime SUHII. The role of waterbodies is significantly positive for the daytime SUHII of Dhaka ($r = 0.886$, $p < 0.01$), Chittagong ($r = 0.823$, $p < 0.01$), Khulna ($r = 0.770$, $p < 0.01$) and Rajshahi ($r = 0.451$, $p < 0.05$) (Table 3). Anthropogenic forcing, defined by Δ NTL, reveals a statistically significant positive relationship with daytime SUHII for four cities, with the exception of Dhaka ($r = -0.443$). The correlation between Δ AOD and daytime SUHII seems to vary depending on the city examined. Both Chittagong and Khulna show significantly positive

relationships between SUHII and albedo (Δ BSA) while the other three cities have insignificantly negative correlations (Table 3). Conversely, nighttime SUHII has a significantly positive relationship with Δ IBI in Rajshahi ($r = 0.524$, $p < 0.05$) while Dhaka and Chittagong both show insignificant correlations ($r = 0.264$; $r = 0.182$, respectively). Population seems to have a significantly positive relationship with nighttime SUHII in Dhaka ($r = 0.546$, $p < 0.05$) and Rajshahi ($r = 0.535$, $p < 0.05$). Greenness shows insignificant negative relationship with nighttime SUHII for four cities, except Khulna ($r = 0.439$) (Table 3). Waterbodies, on the other hand, display a strong relationship with SUHII in four cities. Supplementary Tables (T1 and T2) provide a summary of the relationship between SUHII and driving factors during the four satellite overpass times.

Fig. 5a shows greenness (Δ FVG) has a negative association with day and nighttime temperature, though the strength of the relationship does vary between seasons. On the other hand, Fig. 5 (b) shows a significant relationship between SUHII and imperviousness (Δ IBI) during the day across seasons but at night the relationship varies between cities.

Population shows a significant positive daytime association with SUHII in Dhaka and Chittagong when compared with other three cities, although the relationship is much stronger during winter nights (Fig. 6). Supplementary Table 3 shows the relationship between SUHII and anthropogenic activities (Δ NTL), aerosol (Δ AOD), albedo (Δ BSA) and waterbodies (Δ NDWI) between day and night across cities.

3.4. Trend assessment

Table 4 presents the diurnal SUHII trends produced by the MK test. Results indicate that daytime SUHII appears to have an insignificantly increasing trend for Dhaka, Chittagong and Khulna. Rajshahi displays a statistically significant positive trend only at 1330 (Table 4). Although the nighttime trends (2230 and 0130) are mostly positive (with the exception of Khulna), the values are mostly insignificant (with the exception of Rajshahi at 0130).

An analysis of day and nighttime trends across all seasons indicates that during the pre-monsoon season daytime SUHII shows a significantly increasing trend for Dhaka, Chittagong, Khulna and Sylhet. The nighttime values reveal an insignificantly increasing trend for all cities. A similar overall result can be seen for winter daytime; however, Khulna and Sylhet do show a significantly decreasing trend, and two cities (Dhaka and Rajshahi) reveal a significantly increasing SUHII trend during winter night. In the monsoon season, daytime SUHII increases, though Dhaka and Khulna both record a statistically significant positive trend. During the monsoon night, the SUHII trend exhibits an insignificantly decreasing trend for Chittagong, Khulna, but an insignificantly increasing trend for Dhaka, Rajshahi and Sylhet. A significantly

Table 2
Seasonal SUHII (°C) during satellite overpass times in five cities.

Period	LT	Sensor	Dhaka	Chittagong	Khulna	Rajshahi	Sylhet
Pre-monsoon	1030	Terra	3.13	1.82	1.21	1.35	0.43
	1330	Aqua	3.74	2.14	0.82	1.91	0.51
	2230	Terra	1.63	1.58	0.68	0.77	0.36
	0130	Aqua	1.38	1.44	0.66	0.65	0.51
Monsoon	1030	Terra	3.19	3.15	1.57	1.33	1.73
	1330	Aqua	2.91	2.49	0.53	1.27	0.75
	2230	Terra	0.32	0.94	−0.32	0.40	−0.77
	0130	Aqua	0.28	2.14	−0.14	0.16	−0.77
Post-monsoon	1030	Terra	2.65	2.73	1.40	0.58	1.23
	1330	Aqua	2.70	2.70	0.90	0.58	0.58
	2230	Terra	1.48	1.93	−0.10	0.55	−0.04
	0130	Aqua	1.44	1.75	−0.05	0.42	−0.01
Winter	1030	Terra	2.49	1.43	1.20	0.30	0.74
	1330	Aqua	2.41	1.49	0.89	0.25	0.49
	2230	Terra	2.12	2.32	0.87	0.59	0.75
	0130	Aqua	1.99	2.32	0.83	0.43	0.86

LT: local time.

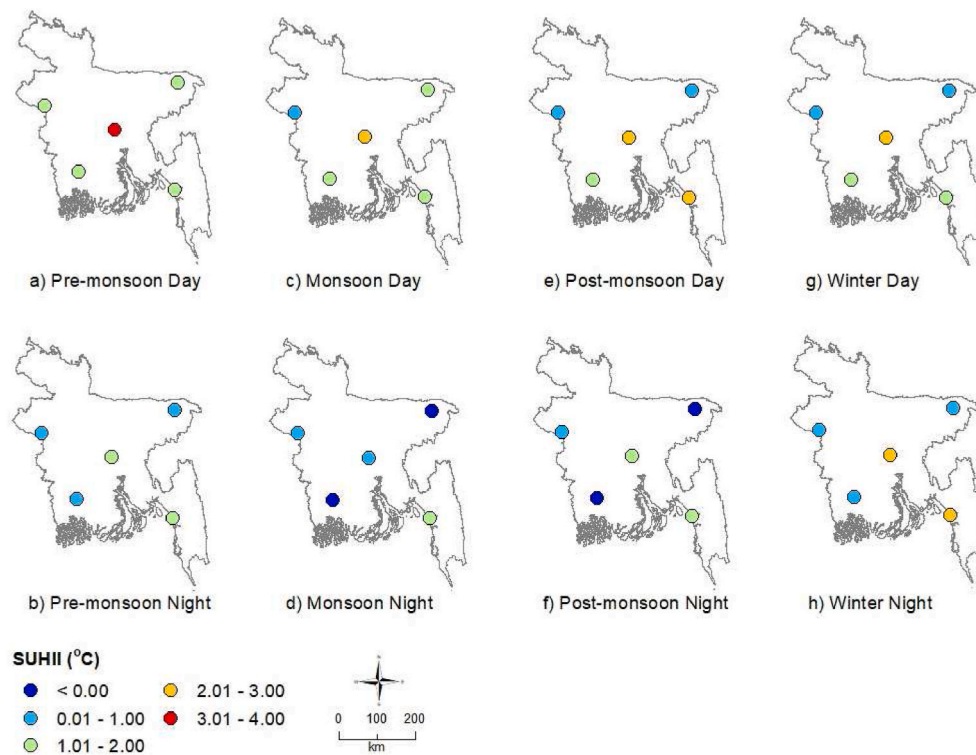


Fig. 4. Spatial distribution of seasonal SUHII (°C) in five cities, averaged over the period of 2003–2019.

Table 3

Relationship between SUHII and driving factors during day and night.

Factors	Dhaka		Chittagong		Khulna		Rajshahi		Sylhet	
	Day	Night	Day	Night	Day	Night	Day	Night	Day	Night
ΔAOD	0.015	0.030	0.135	0.249	−0.000	0.049	0.526**	0.262	0.047	0.120
ΔBSA	−0.414	−0.415	0.551**	0.392	0.520**	−0.856*	−0.102	−0.207	−0.011	−0.364
ΔFVG	−0.891*	−0.526**	−0.868*	−0.118	−0.825*	0.564*	−0.521**	−0.663*	−0.441	−0.342
ΔIBI	0.828*	0.264	0.939*	0.119	0.819*	−0.732*	0.627*	0.524**	0.653*	−0.209
ΔPOP	0.935*	0.546**	0.784*	0.182	0.286	−0.160	0.599*	0.535**	0.486**	0.110
ΔNDWI	0.886*	0.542**	0.823*	0.111	0.770*	−0.448	0.451**	0.678*	0.330	0.377
ΔNTL	−0.443	−0.172	0.739*	0.199	0.495**	−0.639*	0.570**	0.500**	0.566*	0.022

* Significant at 99%; ** Significant at 95% level.

increasing trend is also observed for three cities (Dhaka, Chittagong and Khulna) during post-monsoon daytime, and in the nighttime of the same season an insignificant increasing trend can be found for four cities, apart from Rajshahi which shows a significantly increasing nighttime trend (Fig. 7).

4. Discussion

Unplanned urban expansion, driven by an ever-increasing population resulting from extensive rural-urban migration, is a major issue in the urbanisation process in Bangladesh and has caused significant changes in temperature of large cities (Kant et al., 2018; Roy et al., 2020; Dewan & Corner, 2012). Any increase in existing local urban temperature regimes will increase the susceptibility of citizens to projected heatwave impacts.

The magnitude and causal factors of UHIs vary greatly between cities and climatic zones (Mirzaei & Haghighat, 2010) so using any general conclusions derived from global studies may not be optimal and may lead to uncertainty in regards factors operating at local and regional scales (Manoli et al., 2019). City-specific information based on diurnal and seasonal UHIs and local attributes, is therefore, very important in supporting local scale adaptation and mitigation efforts (Alexander,

2021; Chun & Guldman, 2018).

The observation that the intensity of heating during daytime is greater than during nighttime is consistent with existing studies (Raj et al., 2020; Chakraborty & Lee, 2019; Zhou et al., 2014; Clinton & Gong, 2013; Peng et al., 2012), however the magnitude differs both seasonally and diurnally (Table 2, Figs. 3 and 4). Both under and over-estimation of SUHII are evident when comparing this study to other studies. For example, Chakraborty and Lee (2019) reported that annual average day and nighttime intensity are 1.16 °C and 1.03 °C for Dhaka but this study estimated 2.88 °C and 1.55 °C. Seasonal values were also inconsistent in terms of magnitude, but both studies noted that the summer intensity was larger than winter (Table 2, Supplementary Fig. S3). These differences may have stemmed from: i) the data and methods used (Yao et al., 2018); ii) the LST pixel quality flag used (Lai et al., 2018) and; iii) the definition of a rural site (Manoli et al., 2019). It is reasonable, therefore, to speculate that the data and methods used in the analysis do impact on results and will provide differing estimates of SUHII.

It was also found that the amplitude of day-night temperature changed significantly within Bangladesh cities (Fig. 3 and Supplementary Fig. S3). This finding is consistent with other work (Sun et al., 2019; Huang et al., 2016; Zhao et al., 2014). A narrowing of diurnal amplitude

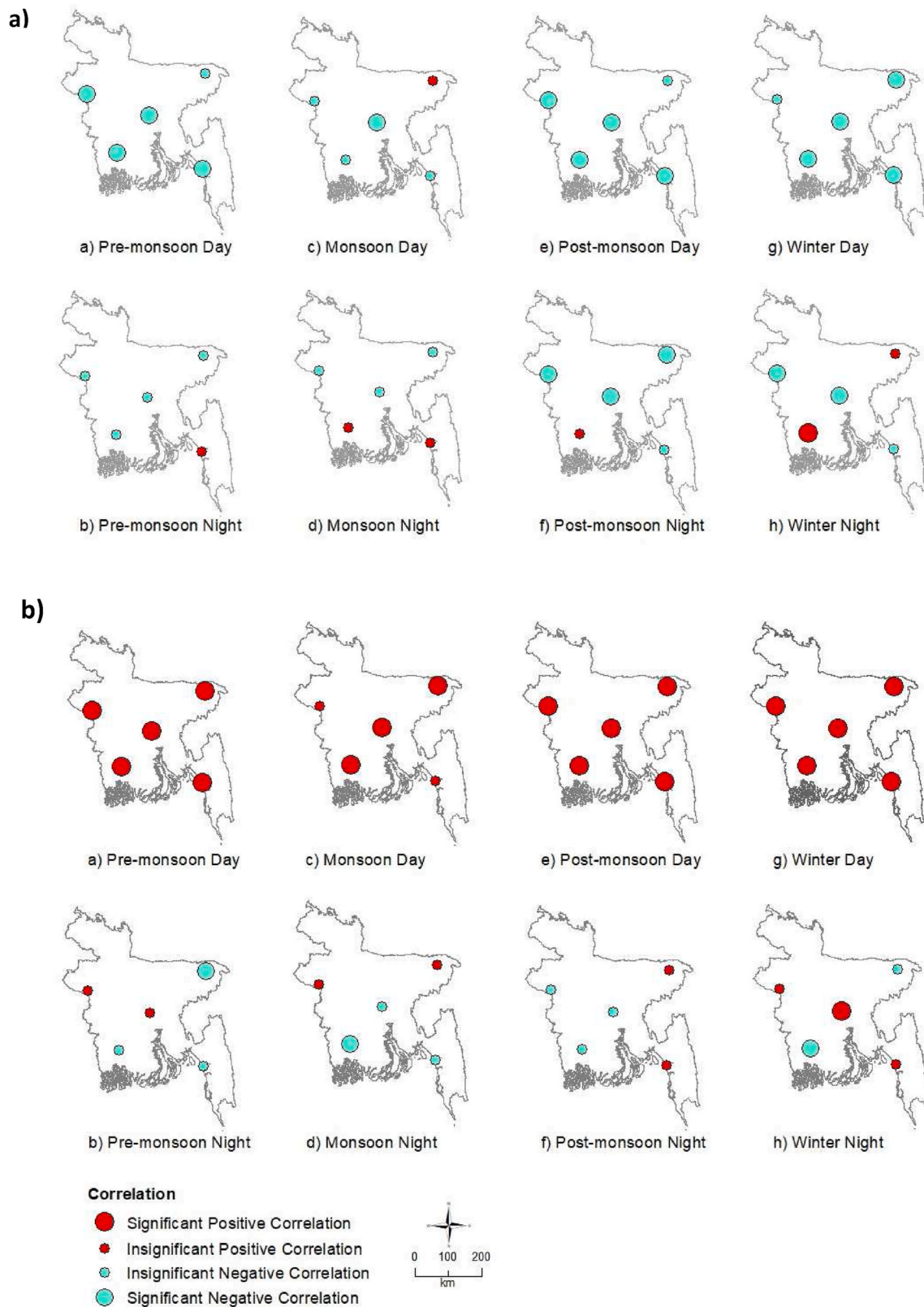


Fig. 5. Pearson's correlation between SUHII and Δ FVG (a); Δ IBI (b).

was more pronounced in Chittagong and Rajshahi, but a broadening was observed in Dhaka. Seasonally, the monsoonal period had a larger day-night amplitude than the pre- and post-monsoon (Fig. 4 and Supplementary Fig. S6), possibly associated with the specific background climate (Sun et al., 2019; Zhao et al., 2014). During the winter season,

however, both Dhaka and Khulna exhibited a positive diurnal amplitude (0.4°C and 0.2°C), while the other three cities had a negative amplitude, with Chittagong recording the largest negative value (-0.86°C). High daytime and strong nighttime SUHII can result in positive and negative diurnal amplitudes in temperature, as observed during the

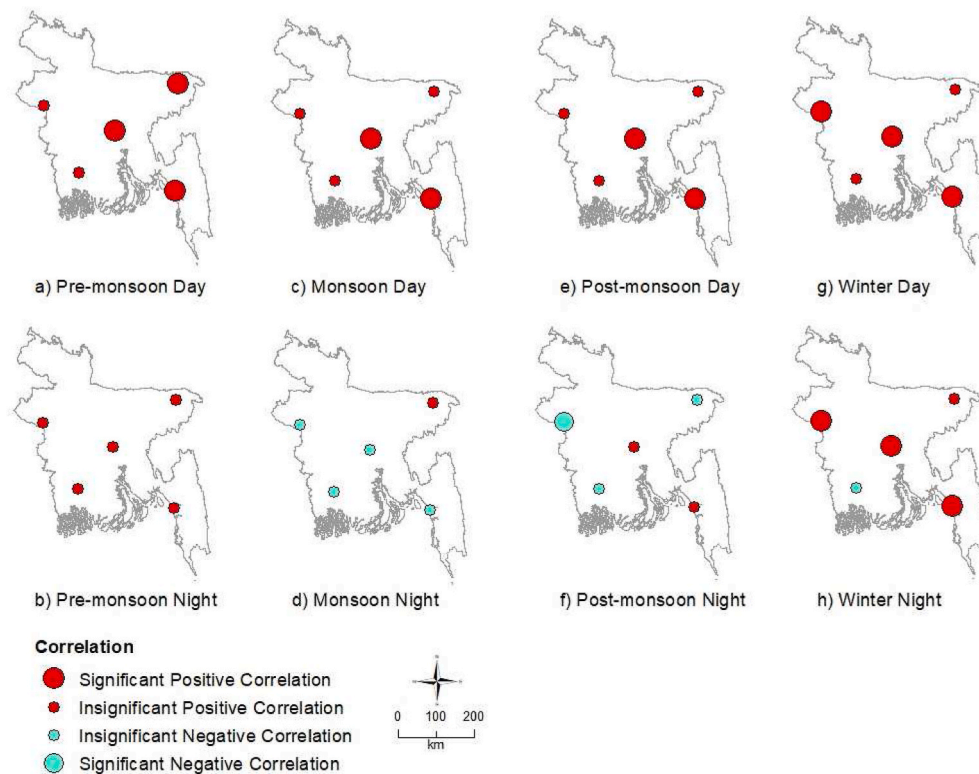


Fig. 6. Pearson's correlation between Δ POP and SUHII during the four seasons in five cities.

Table 4

Trends of SUHII during satellite overpass times and combined datasets ($^{\circ}\text{C}/\text{y}$).

City	Satellite overpass times				Terra plus Aqua	
	1030	1330	2230	0130	Day	Night
Dhaka	0.062*	0.058*	0.014	0.009	0.055*	0.011
Chittagong	0.051*	0.052*	0.005	0.006	0.062*	0.005
Khulna	0.043*	0.045*	-0.017	-0.017*	0.044*	-0.013*
Rajshahi	0.016	0.027*	0.01	0.03*	0.013	0.021*
Sylhet	0.028	0.02	0.004	-0.003	0.020	0.008

* Significant at 95%.

winter season in this study. A similar observation was noted for Beijing and Shanghai (Huang et al., 2016).

Lack of greenness, the amount of imperviousness and population size appeared to be the major determinants, with the role of other drivers such as waterbodies and anthropogenic forcing being inconsistent and difficult to determine. The roles also seemed to vary between cities and seasons (Table 3, supplementary T1-T3). For example, day and night-time SUHI intensity was negatively associated with greenness in all cities, except for a few cases at the seasonal scale (Fig. 5a). This observation reinforces the role of vegetation in mitigating urban warming across cities (Raj et al., 2020; Chakraborty & Lee, 2019; Chen et al., 2017; Zhou et al., 2014; Quan et al., 2016) as greenery reduces thermal load by preventing the warming of air (Bowler et al., 2010). The very dispersed distribution of existing vegetation, however, provides little cooling benefit (Quan et al., 2014). As per capita greenery is very low (Jaman et al., 2020), it is reasonable to state that lack of greenness is probably the single most important factor in intensifying urban warming in Bangladesh. Kant et al. (2018) also observed the same with a Landsat-based study. The planning of vegetation patch locations in areas exposed to urban heating to ensure optimum placement, is therefore, important (Zölch et al., 2016).

Daytime heating, as defined by the urban-rural differences, was significantly associated with the impervious surface (Δ IBI), although the

relationship varied at night. This result aligns with other observations (Raj et al., 2020; Sun et al., 2019; Song & Wu, 2016; Clinton & Gong, 2013; Lazzarini et al., 2013) but contradicts that seen in a study by Feng et al. (2013) who noted little impact of urban growth on surface UHI. Nevertheless, the abundance of impervious surface (in the form of increased numbers of buildings, streets and other anthropogenic features) does have an impact on the energy balance of cities (Oke, 1982). These man-made features not only decrease the amount of solar radiation during daytime, but also reduce the cooling efficiency of long-wave radiation at night due to multiple reflections off urban structures (Wang et al., 2017). The rate of urbanisation currently being experienced by cities in Bangladesh is very high (and mostly in an unplanned manner) so the implementation of city planning policies which specify differing building densities and sparse low-rise structures (Manoli et al., 2019) may play a part in mitigating possible warming effects by changing the energy balance of these urban areas (Mirzaei & Haghighat, 2010).

Despite global studies showing little evidence of any relationship between population and SUHII (Clinton & Gong, 2013; Peng et al., 2012), this study suggested that population size does have a notable impact on surface UHI, particularly in the biggest cities such as Dhaka and Chittagong. This contradicts research by Peng et al. (2012) who reported that city size has little impact on surface UHI intensity. Other studies, ranging from global to local, have also emphasised the impact of population in modulating SUHI (Raj et al., 2020; Manoli et al., 2019; Zhou et al., 2018; Yao et al., 2017; Hung et al., 2006). The cities of Bangladesh are extremely crowded (Rahman et al., 2019), so it is possible to infer that increased socioeconomic activity and greater emissions of anthropogenic heat may explain any spatiotemporal variations in SUHII, as well as the relationships observed with the variables of interest. Although strong correlations between anthropogenic forcing (defined by Δ NTL) and SUHII would be expected, as demonstrated in India by Raj et al. (2020), this study identified mostly negative relationships at both seasonal and diurnal scales for Dhaka. Other factors such as urban form and city shape may also be at play (Zhou et al., 2017) as the urban morphology of Dhaka is extremely dense (Mourshed, 2011). It is interesting that anthropogenic

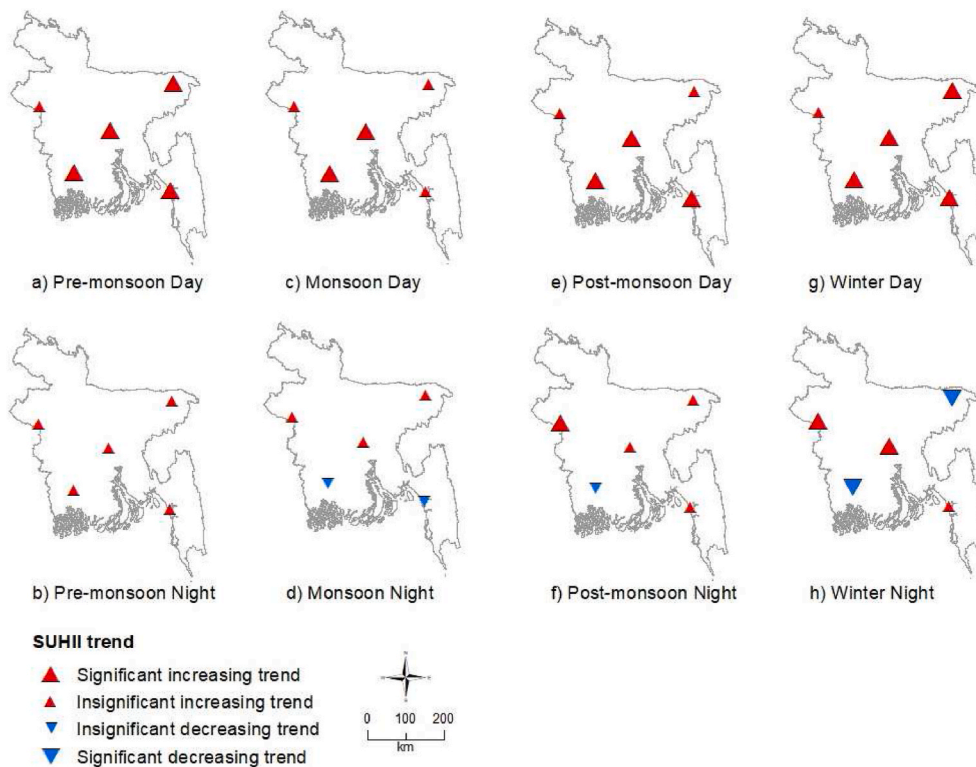


Fig. 7. Seasonal temporal trends of surface urban heat island intensity ($^{\circ}\text{C}/\text{y}$) in each city from 2003 to 2019.

activities seem to have an important role to play in modifying the SUHII of other cities (both during the day and at night) despite the observed negative relationship seen during monsoonal and winter nights in Khulna and Rajshahi (Table 3, supplementary T3). This observation is generally consistent with Zhou et al. (2014).

Although SUHII was significantly influenced by waterbodies at the diurnal scale, the impact appeared to diminish at a seasonal scale, particularly in Dhaka (Table 2, supplementary T1-T3). Waterbodies displayed a statistically significant relationship with daytime heating in Chittagong across all seasons, but had a weak association at night (supplementary T3). While daytime SUHII generally indicated an association with water features in Khulna, Rajshahi and Sylhet during all four seasons, the relationship was negative during winter night (Supplementary T3). These cities are located beside large waterbodies, so land-sea or land-river breezes may have an impact on SUHI intensity (Gunawardena et al., 2017). Song and Wu (2016) noted that the movement of hot air from a warmer surface can heat up adjacent waterbodies, so it is possible that waters can increase in temperature when in contact with warm urban areas. This juxtaposition was seen in the current study.

The current analysis could be improved by incorporating a number of other factors. These include background climate, the vertical and horizontal dimensions of urban morphology, landscape composition and configuration, and waste heat generation. Satellite-based LST measurements tend to overestimate the temperature of horizontal urban features and underestimate surface temperature of vertical features (Hu et al., 2016), so detailed (in-field) monitoring of city microclimates is urgently needed. As the urban population of Bangladesh is projected to be 81.4 million in 2029 (BBS, 2017), any further rise in urban temperatures will have serious implications for these most densely populated cities. This study provided insights into the spatiotemporal intensity of SUHI and associated determinants and temporal trends. This information can be used to develop area-specific strategies for mitigation and adaptation

efforts to combat heatwave conditions which are projected to occur with increasing frequency in the South Asia region (Im et al., 2017).

5. Conclusion

This study examined the diurnal and seasonal intensity of SUHI in five large cities of Bangladesh. Quality-controlled data, recorded by the MODIS Terra and Aqua satellites for the 2003 to 2019 time period, was used. Biophysical and socioeconomic factors were correlated with SUHII and the temporal trend was examined via the Mann-Kendall test. Results indicated that the SUHI intensity differed significantly between day and night, though the magnitude tended to be larger during the day than at night. Diurnal and seasonal amplitude of surface temperature were characterised by narrowing and broadening, especially in the winter season.

An analysis of correlations between drivers and SUHII indicated that lack of greenness, the amount of impervious surface, and population were major factors influencing SUHII in the five cities, particularly during the day. The daytime increasing trend of SUHII was more pronounced than that noted at night, and varied both diurnally and seasonally.

While public policies in Bangladesh are well developed in regards the natural hazards that frequently strike the country, the phenomenon of urban warming has not received due attention in any of the current, major policy documents. These include the Bangladesh Climate Change Strategy and Action Plan 2009 (MoEF, 2009), and Seven Five Year Plan (GED, 2016). The Bangladesh Delta Development Plan (BDP2100) does recognise the urban heat island effect as an issue though the specific actions documented are not well conceived (GED, 2018). Bangladesh is frequently named as one of the world's most vulnerable countries in respect to climate change effects. The current study underlines the fact noted in other research that urban climate adaptation plans need to be prioritised to generate local 'climate-smart' strategies (Sun et al., 2019).

CRediT authorship contribution statement

Ashraf Dewan: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Grigory Kiselev:** Data curation, Formal analysis. **Dirk Botje:** Data preparation, Writing – review & editing, and analysis. All the authors have read, reviewed and approved the submitted manuscript.

Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apgeog.2021.102533>.

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